

Asian Transport Database Overview

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1 Literature review

Completed a literature review to understand datasets/ complete work already published and open source.

Outputs:

Asia Transport Literature Tracker.xlsx

- Includes relevant publications, reports, datasets, organisations, list of countries in study area and modes relevant to each country.

Asia Transport Literature Tracker.pdf

- Summary of spreadsheet

2 Airports

2.1 Data sources used

2.2 Analysis done

Airport node spatial data correction (completed in Airport.py)

The World Bank airport dataset provided the only identified open source data containing flight connection paths between airports. Therefore, the World Bank airport points were used as the basis for the airport nodes used for the flight network.

However, the World Bank airport coordinates were found to have poor spatial accuracy. The airport locations were therefore checked and corrected using the OurAirports dataset, which appeared to provide better spatial accuracy.

The analysis used airport data from Ourairports.csv, worldbank_filtered_airport_flows.gpkg and worldbank_filtered_airport_volume.gpkg.

OurAirports data was first filtered to include only countries within the study area using Countries_list.xlsx. The World Bank airport coordinates were then corrected using the OurAirports coordinates where a matching IATA code was available. Where no matching OurAirports record was available, the original World Bank coordinates were retained.

The origin and destination coordinates of the World Bank flight connections were then corrected using the corrected airport coordinates.

airport_coordinate_audit.xlsx contains the audit of this matching process. The match_summary sheet provides a summary of the number of airport locations that were common between the two datasets. 14 airports in the World Bank dataset could not be matched and therefore could not be spatially corrected using OurAirports.

2.2.1 Shapefile matching (OSM_extract_country_loop and airport_shapefile_nearest.py):

Relevant tags were extracted from OSM.

For Aiports these were:

```
{"aeroway": ["aerodrome"]},
```

```
{"aeroway": ["terminal"]},
```

```
{"aeroway": ["runway"]},
```

```
{"aeroway": ["taxiway"]},
```

From here, any extracted OSM features located within 2 km of an airport node in `world-bank_filtered_airport_volume.gpkg` were assigned to that airport. The airport's IATA code was then added to the feature's attributes. The final output combines all extracted OSM airport features into a single layer, with attributes identifying the feature's original OSM tag (e.g. `aerodrome`, `terminal`, `runway` or `taxiway`) and the airport it was assigned to.

2.2.2 Still to do:

- Check the spatial location of the 14 airport nodes in `airport_flows_world_bank_corrected.gpkg`, which did not have a match to the OurAirports database. (Validating that the flows also update if any airports get updated)
- Add missing polygon data to the XX airports

3 Maritime

3.1 Data sources used/outputted

- Port watch
- Multi-hazard risk to global port infrastructure and resulting trade and logistics losses, [Jasper Verschuur](<https://www-nature-com.ezproxy-prd.bodleian.ox.ac.uk/articles/s43247-022-00656-7#auth-Jasper-Verschuur-Aff1>)
- Port_landuse.gpkg (shapefiles for ports)
- Global_maritime_network.gpkg (each node is at the centre of the port_landuse shapefiles that it is relevant to)
- Ports.shp

3.2 Analysis done

3.2.1 Maritime network (completed in `ports.py`):

Code initially copied from the `African-transport-dataset/scripts/preprocess/"ports_new_merge.py"`. Using the `Ports.shp` and updated 2023-2025 attribute data provided by Jasper Verschuur (`port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`, `port_calls_average_2023-2025.csv`). A node and edge network was created, with the attributes. The network was broken up into 3 clusters, to aid visuals (`northern_russia`, `pacific_network` and `large_cluster`). The output contains two node types, ports (land) and maritime (networking), alongside the edge connections between.

-file output names

`global_maritime_network_PROVA_NEW1`

`asia & pacific_maritime_network_PROVA_NEW1`

`northern_russia_network_maritime_network`

`large_cluster_maritime_network`

`pacific_network_maritime_network`

3.2.2 Shapefile correction (all done using `port_shapefile_nearest.py` and manually):

`port_landuse.gpkg` shapefiles were matched to their corresponding ports by assigning the nearest port node to each polygon (nodes stored in `asia_pacific_maritime_network_PROVA_NEW1.gpkg`). The results of this automated matching are stored in `port_landuse_nearest_port_id_lookup.xlsx`.

The accuracy of these matches was then manually checked and recorded in `port_landuse_nearest_port_id_lookup.xlsx`. Each match was assessed using the distance between the port node and the land-use polygon, the names associated with both datasets, and the port's location on Google Maps. In the checked file, the checked column indicates whether the port land-use shapefile (Column A) has been matched to the correct port node (Column H). Where checked indicates that the match is incorrect, the correct port id column records the correct port ID, identified manually.

The corrections were then applied, and the `port_landuse_nearest_port_ids.gpkg` and `port_landuse_nearest_port_ids.xlsx` contain the correct results.

14 ports from the `asia_pacific_maritime_network_PROVA_NEW1.gpkg` file did not have any polygon data (from `port_landuse.gpkg`) matched to it. Seen in `port_landuse_nearest_port_id_lookup_checked.xlsx` sheet-node_polygon_audit, column `missing_polygon_data = TRUE`.

3.2.3 Still to do:

- Add missing polygon data to the 14 ports.

4 Road

5 Rail

6 Inland waterways

Investigated OSM tags, best appeared to be:

Tag = `boat:yes`

Tag = `ferry` : terminal

7 Multimodal

8 ARC

ARC is separated into two file directories:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/asia-transport-database
```

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

9 Using open-gira on ARC

Purpose: Used to create the road and rail networks using OpenGIRA.

1. Go to the OpenGIRA directory:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/open-gira
```

2. Activate the OpenGIRA environment:

```

module load Mamba/23.11.0-0
export CONDA_ENVS_PATH=/home/mans4968/.mamba/envs
export MAMBA_ROOT_PREFIX=/home/mans4968/.mamba
eval "$(conda shell.bash hook)"
conda activate /home/mans4968/.mamba/envs/open-gira

```

3. Run OpenGIRA using Snakemake and the ARC profile (not run on a .sh file):

```
snakemake -profile config/ARC -cores 1
```

A specific output can also be requested, for example:

```
snakemake -profile config/ARC -cores 1 results/russia-latest_filter-road-primary/edges.
```

4. Check jobs currently running:

```
squeue -u $USER
```

5. Check completed/failed jobs:

```
sacct -u $USER -format=JobID,JobName,State,Elapsed,ExitCode
```

6. Outputs are stored in:

```
/data/cenv-opsis-astdb/{<USERNAME>}/open-gira/results/
```

Completed final networks are organised into:

```
results/road/
```

```
results/rail/
```

Other folders such as geoparquet/, slices/, json/ and input/ contain input or intermediate OpenGIRA data.

10 Asia-transport-database on ARC

Purpose: Used to run the Asia transport database processing, including the country-by-country OSM infrastructure extraction.

1. Go to the Asia database:

```
cd /data/cenv-opsis-astdb/{<USERNAME>}/asia-transport-database
```

2. Input OSM data are stored in:

```
incoming_data/osm/
```

3. Run the required Asia transport database script/job on ARC.

See various .sh files

4. Processed outputs are stored in:

```
processed_data/infrastructure/
```

11 Already existing country multimodal transport packages

Multimodal transport data packages have already been extracted for China, Singapore, New Zealand, and Australia. These datasets can be used later for comparison and validation against the wider transport database. The packages are stored under:

`incoming_data/infrastructure/[folder name]`

The content of each country package varies, and not all packages contain data for every transport mode.

12 Files Overview

Asia-transport-database repository

A catalogue of inputs, intermediate files, final outputs and processing scripts

Legend: entries marked with a blue star (★) are final outputs.

Files are grouped by their repository location; the descriptions below record their purpose, provenance and producing script.

1. `scripts/preprocess/Airport.py`

Purpose:

Filters OurAirports to study countries, corrects World Bank airport coordinates by IATA code, and updates airport flow coordinates.

Key inputs:

`incoming_data/infrastructure/Airport/Ourairports.csv`; `incoming_data/infrastructure/Airport/worldbank_filtered_airports.csv`; `incoming_data/infrastructure/Airport/worldbank_filtered_airport_flows.gpkg`; `incoming_data/Countries_list.xlsx`

Additional related input present:

`incoming_data/infrastructure/Airport/airport_coordinate_audit_corrected.xlsx`

Key outputs:

`filtered_Ourairports.csv`; `filtered_Ourairports.gpkg`; `\{*world_bank_airports_corrected.gpkg`; `\{*airport_flows_world_bank_corrected.gpkg`; `airport_coordinate_audit.xlsx`; `airport_coordinate_audit_applied.xlsx`

2. `scripts/preprocess/airport_shapefile_nearest.py`

Purpose:

Matches OSM airport lines/polygons to corrected airport points within 2 km and produces audit summaries.

Key inputs:

`processed_data/infrastructure/airport/world_bank_airports_corrected.gpkg`; `processed_data/infrastructure/osm_filter/filtered_osm_features_within_2km.gpkg`

Key outputs:

`\{*airport_osm_features_within_2km.gpkg`; `airport_osm_features_within_2km.xlsx`

3. `scripts/preprocess/OSM_extract_country_loop.py`

Purpose:

Extracts selected OSM tags by country/region from source PBF files.

Key inputs:

incoming_data/osm/[country/region].osm.pbf; incoming_data/Countries_list.xlsx

Key outputs:

processed_data/infrastructure/osm_filter/[country/region]_[tag].parquet

4. scripts/preprocess/ports.py

Purpose:

Builds global and Asia-Pacific maritime network node/edge datasets, including clustered outputs.

Key inputs:

incoming_data/Global port supply-chains/Ports Updated 2025/Ports.csv; incoming_data/Global port supply-chains/Ports Updated 2026/port_calls_average_2023-2025.csv; incoming_data/Global port supply-chains/Ports Updated 2026/port_capacity_called_average_2023-2025.csv; incoming_data/Global port supply-chains/Ports Updated 2026/port_turn_around_time_average_2023-2025.csv; incoming_data/Countries_list.xlsx

Additional source files present (same port source package):

Ports.shp; Ports.shx; Ports.dbf; Ports.prj; Ports.cpg; Ports.qmd

Key outputs:

global_maritime_network_PROVA_NEW1.gpkg; global_maritime_network.gpkg; asia_pacific_maritime_network_PROVA_NEW1.gpkg; northern_russia_network_maritime_network.gpkg; large_cluster_maritime_network.gpkg; pacific_network_maritime_network.gpkg; \{*cleaned_asia_pacific_maritime_network_PROVA_NEW1.gpkg

Additional output directory present:

processed_data/infrastructure/port/Seperate maritime nodes and edges/

5. scripts/preprocess/port_shapefile_nearest.py

Purpose:

Matches port land-use polygons to nearest maritime network nodes, then applies manual corrections.

Key inputs:

incoming_data/infrastructure/port/port_landuse.gpkg; incoming_data/infrastructure/port/port_landuse_nearest_port.gpkg; processed_data/infrastructure/port/asia_pacific_maritime_network_PROVA_NEW1.gpkg

Key outputs:

\{*port_landuse_nearest_port_ids.gpkg; port_landuse_nearest_port_id_lookup.xlsx; port_landuse_nearest_port_id_lookup.gpkg

6. scripts/preprocess/Combine_opengira.py

Purpose:

Combines and deduplicates overlapping OpenGIRA node/edge extracts into study-area mode-specific outputs.

Key inputs:

processed_data/open_gira/results/[transport_mode]/[country/region]_*/edges.gpq; processed_data/open_gira/results/[transport_mode]/[country/region]_*/nodes.gpq

Key outputs:

```
\{*processed_data/open_gira/results/rail/combined/asia_pacific_rail_edges.gpq; \{*processed_data/open_gira/results/road_primary/combined/asia_pacific_road_primary_edges.gpq; \{*processed_data/open_gira/results/road_primary/combined/asia_pacific_road_primary_nodes.gpq; [transport_mode]_kept_summary.xlsx (if audit export enabled)
```

7. scripts/preprocess/utils_new.py

Purpose:

Shared helper functions used by preprocessing scripts; copied/adapted from African-transport-dataset/scripts/preprocess/.

8. Shared external references used across preprocessing

<https://download.geofabrik.de/index.html> (OSM PBF); <https://ourairports.com/data/> (OurAirports); <https://datacatalog.worldbank.org/search/dataset/0038117/global-airports> (World Bank airports); <https://portwatch.imf.org/> (PortWatch attributes); <https://data.mendeley.com/datasets/kdyt24tsh5/1> (Port spatial package)